Exploring Distributions

STA 310: Homework 2

Instructions

- Write all narrative using full sentences. Write all interpretations and conclusions in the context of the data.
- Be sure all analysis code is displayed in the rendered pdf.
- If you are fitting a model, display the model output in a neatly formatted table. (The `tidy` and `kable` functions can help!)
- If you are creating a plot, use clear and informative labels and titles.
- Render and back up your work reguarly, such as using Github.
- When you're done, we should be able to render the final version of the Rmd document to fully reproduce your pdf.
- Upload your pdf to Gradescope. Upload your Rmd, pdf (and any data) to Canvas.

These exercises come from BMLR or are adapted from BMLR, Chapter 3.

Exercises

Exercise 1

At what value of p is the variance of a **binary random variable** smallest? When is the variance the largest? Back up your answer empirically or mathematically.

Mathematical Proof

Let $X \sim \text{Bernoulli}(p)$, where $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

The variance of a Bernoulli random variable is

$$\text{Var}(X) = p(1 - p).$$

Rewrite this expression as a quadratic in p:

$$\text{Var}(X) = p - p^2 = -\left(p^2 - p\right).$$

Complete the square:

$$\text{Var}(X) = -\left[\left(p - \tfrac{1}{2}\right)^2 - \tfrac{1}{4}\right] = -\left(p - \tfrac{1}{2}\right)^2 + \tfrac{1}{4}.$$

Since $\left(p - \frac{1}{2}\right)^2 \geq 0$ for all $p \in [0, 1]$, it follows that

$$\text{Var}(X) \leq \tfrac{1}{4}.$$

Equality holds when

$$p = \tfrac{1}{2},$$

so the variance is maximized at $p = \frac{1}{2}$ with

$$\text{Var}(X) = \tfrac{1}{4}.$$

To find the minimum, note that

$$p(1-p) = 0 \iff p = 0 \text{ or } p = 1.$$

Thus, the variance is minimized at $p = 0$ or $p = 1$, where

$$\text{Var}(X) = 0.$$

Empirical Check in R

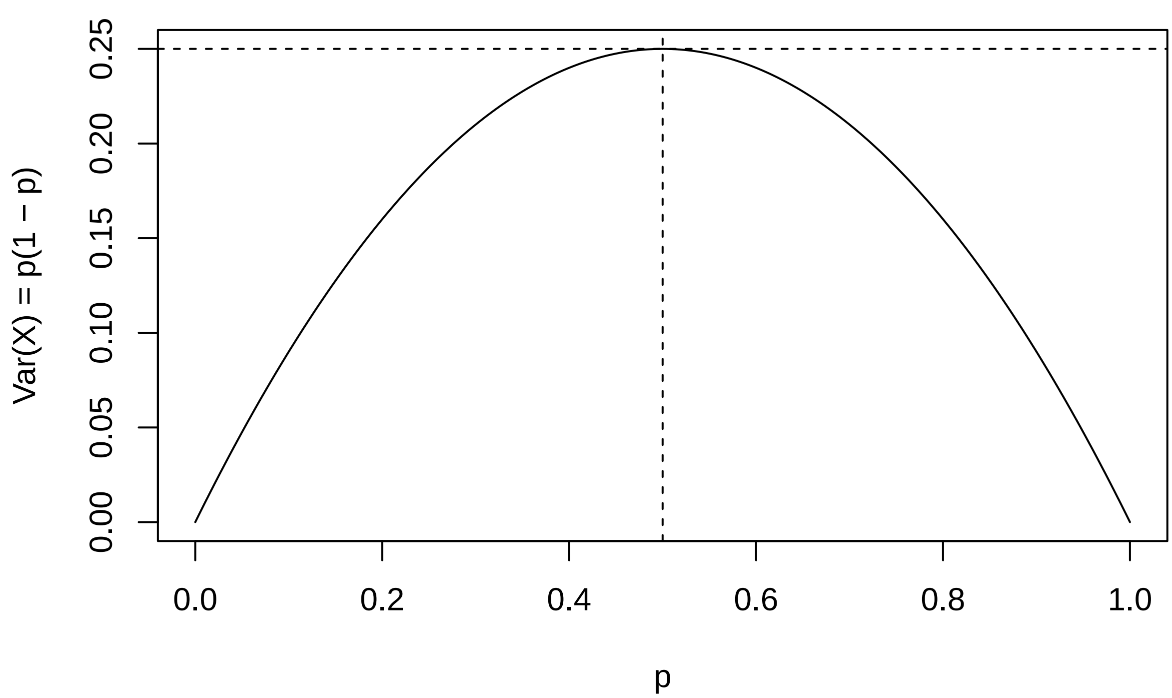
```r
p <- seq(0, 1, by = 0.001)
# Create a sequence of probability values from 0 to 1 in increments of 0.001

v <- p * (1 - p)
# Compute the variance of a Bernoulli(p) random variable for each value of p

plot(p, v, type = "l", xlab = "p", ylab = "Var(X) = p(1 - p)")
# Plot variance as a function of p, using a line plot

abline(v = 0.5, lty = 2)
# Add a vertical dashed line at p = 0.5, where the variance is maximized

abline(h = 0.25, lty = 2)
```



```
# Add a horizontal dashed line at variance = 0.25, the maximum possible variance
```

Exercise 2

How are hypergeometric and binomial random variables different? How are they similar?

Both the binomial and hypergeometric random variables model the number of successes in a fixed number of trials. In each case, every trial results in one of two possible outcomes: success or failure. Consequently, the support of both random variables is the set $\{0, 1, \ldots, n\}$.

For both distributions, the expected number of successes is the same. Let p denote the proportion of successes in the population, for both distributions:

$$\mathbb{E}[X] = np.$$

Thus, on average, the two models predict the same number of successes when the underlying success probability is p.

When the population size is large relative to the sample size, the binomial distribution provides a good approximation to the hypergeometric distribution.

The primary difference of the two distributions is in the sampling mechanism. A binomial random variable assumes that sampling is performed *with replacement*. As a result, the trials are independent and the probability of success remains constant across trials. In contrast, a hypergeometric random variable assumes sampling *without replacement*, which introduces dependence between trials and causes the probability of success to change after each draw.

A binomial random variable is defined as

$$X \sim \text{Binomial}(n, p),$$

with probability mass function

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

A hypergeometric random variable is defined as

$$X \sim \text{Hypergeometric}(N, K, n),$$

where N denotes the population size, K denotes the number of successes in the population, and n denotes the sample size. Its probability mass function is

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Another important distinction between the two distributions is their variances. For a binomial random variable,

$$\text{Var}(X) = np(1-p).$$

For a hypergeometric random variable, the variance is given by

$$\text{Var}(X) = np(1-p)\frac{N-n}{N-1}.$$

The factor $\frac{N-n}{N-1}$, known as the finite population correction, reflects the dependence between trials and causes the hypergeometric variance to be smaller than the binomial variance.

When the population size N is much larger than the sample size n, the finite population correction factor satisfies

$$\frac{N-n}{N-1} \approx 1.$$

In this setting, the hypergeometric distribution is well approximated by a binomial distribution with

$$p = \frac{K}{N}.$$

Exercise 3

How are exponential and Poisson random variables related?

Poisson and exponential random variables arise from the same stochastic process (the Poisson process). A Poisson process models the occurrence of events over time under the assumptions that events occur independently and at a constant average rate.

Let events occur according to a Poisson process with rate $\lambda > 0$. The number of events occurring in a time interval of length t is a Poisson random variable:

$$N(t) \sim \text{Poisson}(\lambda t),$$

with probability mass function

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k = 0, 1, 2, \ldots$$

The waiting time until the first event occurs is an exponential random variable. Let T denote the time to the first event. Then

$$T \sim \text{Exponential}(\lambda),$$

with probability density function

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

The connection between the two distributions can be seen through the event

$$\{T > t\} = \{N(t) = 0\}.$$

That is, the waiting time to the first event exceeds t if and only if no events occur in the interval $[0, t]$. Therefore,

$$P(T > t) = P(N(t) = 0) = e^{-\lambda t},$$

which is exactly the survival function of an exponential random variable with rate λ.

More generally, the waiting time until the r-th event occurs follows a Gamma distribution:

$$T_r \sim \text{Gamma}(r, \lambda),$$

which is the sum of r independent exponential random variables, each with rate λ.

In a Poisson process, the Poisson distribution describes the number of events occurring in a fixed time interval, while the exponential distribution describes the waiting time between consecutive events.

Exercise 4

How are geometric and exponential random variables similar? How are they different?

Geometric and exponential random variables both model waiting times until the first success or event occurs. The geometric distribution measures the number of discrete trials required to obtain the first success, while the exponential distribution measures the amount of continuous time until the first event occurs.

Both distributions are memory-less. For a geometric random variable $X \sim \text{Geometric}(p)$,

$$P(X > m + n \mid X > m) = P(X > n).$$

Similarly, for an exponential random variable $T \sim \text{Exponential}(\lambda)$,

$$P(T > s + t \mid T > s) = P(T > t).$$

The memoryless property implies that the probability of waiting an additional amount of time does not depend on how long one has already waited.

Both distributions are also associated with renewal processes. The geometric distribution arises from repeated Bernoulli trials with constant success probability p, while the exponential distribution arises from a Poisson process with constant rate λ.

The main difference between the two distributions lies in the nature of their supports. A geometric random variable is discrete and takes values in the set

$$\{1, 2, 3, \ldots\},$$

representing the trial on which the first success occurs. An exponential random variable is continuous and takes values in

$$[0, \infty),$$

representing the continuous waiting time until the first event.Their probability models also differ. The probability mass function of a geometric random variable is

$$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \ldots$$

whereas the probability density function of an exponential random variable is

$$f_T(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

The parameters have different interpretations. In the geometric distribution, p represents the probability of success on each trial, while in the exponential distribution, λ represents the rate at which events occur per unit time.

Finally, the geometric distribution can be viewed as a discrete analogue of the exponential distribution. As the trial spacing becomes small and the success probability per trial becomes small, the geometric distribution converges to the exponential distribution.

Exercise 5

A university's college of sciences is electing a new board of 5 members. There are 35 applicants, 10 of which come from the math department. What distribution could be helpful to model the probability of electing X board members from the math department?

The hypergeometric distribution is appropriate for modeling this situation.

A total of 5 board members are selected from a finite population of 35 applicants, 10 of whom are from the math department. Because applicants are selected without replacement, the number of board members chosen from the math department is a hypergeometric random variable.

Let X denote the number of elected board members from the math department. Define

$$N = 35 \quad \text{(total applicants)}, \qquad K = 10 \quad \text{(math department applicants)}, \qquad n = 5 \quad \text{(board members selected)}.$$

Then

$$X \sim \text{Hypergeometric}(N = 35, K = 10, n = 5).$$

The probability mass function of X is

$$P(X = k) = \frac{\binom{10}{k}\binom{25}{5-k}}{\binom{35}{5}}, \quad k = 0, 1, 2, 3, 4, 5.$$

Exercise 6

Chapter 1 asked you to consider a scenario where *"The Minnesota Pollution Control Agency is interested in using traffic volume data to generate predictions of particulate distributions as measured in counts per cubic feet."* What distribution might be useful to model this count per cubic foot? Why?

A Poisson distribution would be useful for modeling the particulate count per cubic foot.

This is because the data consist of counts of particles occurring in a fixed volume of space (one cubic foot). The Poisson distribution is commonly used to model the number of events occurring in a fixed region of time or space when the events occur independently and at a constant average rate.

Let X denote the number of particulate matter particles observed in one cubic foot of air. Then

$$X \sim \text{Poisson}(\lambda),$$

where $\lambda > 0$ represents the average number of particles per cubic foot.

The Poisson distribution is appropriate in this context because particulate particles can be viewed as randomly occurring events in space, individual particles do not influence the occurrence of others, and the expected rate of occurrence can be reasonably assumed to be constant over small spatial regions.

Rather than assuming a single constant rate, the rate parameter can be modeled as a function of traffic volume. For example, if V denotes traffic volume, one could write

$$\lambda = f(V),$$

where $f(\cdot)$ is an increasing function. This could be modeled using a log-linear relationship such as

$$\log(\lambda) = \beta_0 + \beta_1 V,$$

which ensures that λ remains positive and allows traffic volume to systematically influence particulate counts.

Exercise 7

Chapter 1 also asked you to consider a scenario where *"Researchers are attempting to see if socioeconomic status and parental stability are predictive of low birthweight. They classify a low birthweight as below 2500 g, hence our response is binary: 1 for low birthweight, and 0 when the birthweight is not low."* What distribution might be useful to model if a newborn has low birthweight?

Exercise 8

Chapter 1 also asked you to consider a scenario where *"Researchers are interested in how elephant age affects mating patterns among males. In particular, do older elephants have greater mating success, and is there an optimal age for mating among males? Data collected includes, for each elephant, age and number of matings in a given year."* Which distribution would be useful to model the number of matings in a given year for these elephants? Why?

Exercise 9

Describe a scenario which could be modeled using a gamma distribution.

Exercise 10

Beta-binomial distribution. We can generate more distributions by mixing two random variables. Beta-binomial random variables are binomial random variables with fixed n whose parameter p follows a beta distribution with fixed parameters α, β. In more detail, we would first draw p_1 from our beta distribution, and then generate our first observation y_1, a random number of successes from a binomial (n, p_1) distribution. Then, we would generate a new p_2 from our beta distribution, and use a binomial distribution with parameters n, p_2 to generate our second observation y_2. We would continue this process until desired.

Note that all of the observations y_i will be integer values from $0, 1, \ldots, n$. With this in mind, use `rbinom()` to simulate 1,000 observations from a plain old vanilla binomial random variable with $n = 10$ and $p = 0.8$. Plot a histogram of these binomial observations. Then, do the following to generate a beta-binomial distribution:

a. Draw p_i from the beta distribution with $\alpha = 4$ and $\beta = 1$.
b. Generate an observation y_i from a binomial distribution with $n = 10$ and $p = p_i$.
c. Repeat (a) and (b) 1,000 times ($i = 1, \ldots, 1000$).
d. Plot a histogram of these beta-binomial observations.

Compare the histograms of the "plain old" binomial and beta-binomial distributions. How do their shapes, standard deviations, means, possible values, etc. compare?

Grading

Total	**33**
Ex 1	5
Ex 2	5
Ex 3	5
Ex 4	5
Ex 5	2
Ex 6	2
Ex 7	2
Ex 8	2
Ex 9	2
Ex 10	5
Workflow & formatting	3

The "Workflow & formatting" grade is to based on the organization of the assignment write up along with the reproducible workflow. This includes having an organized write up with neat and readable headers, code, and narrative, including properly rendered mathematical notation. It also includes having a reproducible Rmd or Quarto document that can be rendered to reproduce the submitted PDF.